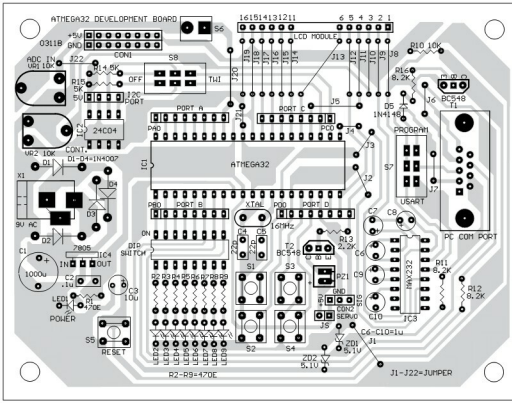



Fig. 4: Component layout for the PCB

to further help in debugging. The buzzer can be disconnected from the circuit at any point of time by removing shorting jumper JS.

In order to further enhance the user interface of the development board, a dedicated LCD port is provided on-board. To minimise use of I/O pins for LCD control, the display is made to function in 4-bit mode. In 4-bit mode, only four data pins are required in addition to RS, RW and E control pins. In total, seven I/O port pins are required for functioning of the display.

Details about the LCD module and its interfacing with the MCU are provided later in the article. Port C is used to interface the LCD in the development board. Port-C pins PC4, PC5, PC6 and PC7 form data pins, whereas PC2 and PC3 are used as RS and E control pins. Port-D pin PD6 is used to provide R/W control signal.

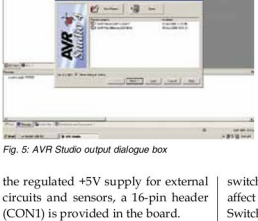


Fig. 5: AVR Studio output dialogue box

the regulated +5V supply for external circuits and sensors, a 16-pin header (CON1) is provided in the board.

The development board has self-contained regulated +5V supply circuit built around the 7805 voltage regulator and filtering capacitors to obtain a filtered, regulated power supply. A power-indicator LED is used to indicate the presence of power supply in the circuit. The power supply circuit with 9V input to 7805 is shown in Fig. 2.

For easy debugging, the board has eight LEDs (LED2 through LED9) selectable through a DIP switch. All these LEDs are connected to Port B of the MCU through current-limiting resistors. When not in use, the LEDs can be disconnected from the circuit individually with the help of DIP switches, thus enabling the normal operation of Port B for other applications.

Five tactile pushbutton switches are also provided for debugging purpose. All the five switches are active-low type and do not affect the normal operation of port pins. Switches S1, S2, S3 and S4 are connected to Port D pins PD0, PD1, PD2 and PD3, respectively. PD0 and PD1 form external interrupt pins of the MCU.

A power-on reset circuit is built around reset pin 9 with the help of resistor R10 and capacitor C3. Switch S5 is connected to pin 9 for resetting the MCU externally.

At Port PD7 (pin 21), a piezobuzzer is connected via switching transistor T2

In order to communicate with the PC, the development board uses the USART module of the MCU. Functional details of this module are given later in the article. The RS-232 signals range between +12V and -12V and are inverted (-12V is a logical 1). Applying this kind of voltage to the AVR is dangerous, so MAXIM's MAX232 level converter is used between the DB9 female serial port and the MCU. The MAX232 uses capacitors to perform the necessary level shifting in order to receive and transmit signals. The same DB9 serial port connector is used to program the MCU by using the SPI serial programming facility available in the MCU.

Popular Open Source software PonyProg is used for easy programming of the MCU. The same software